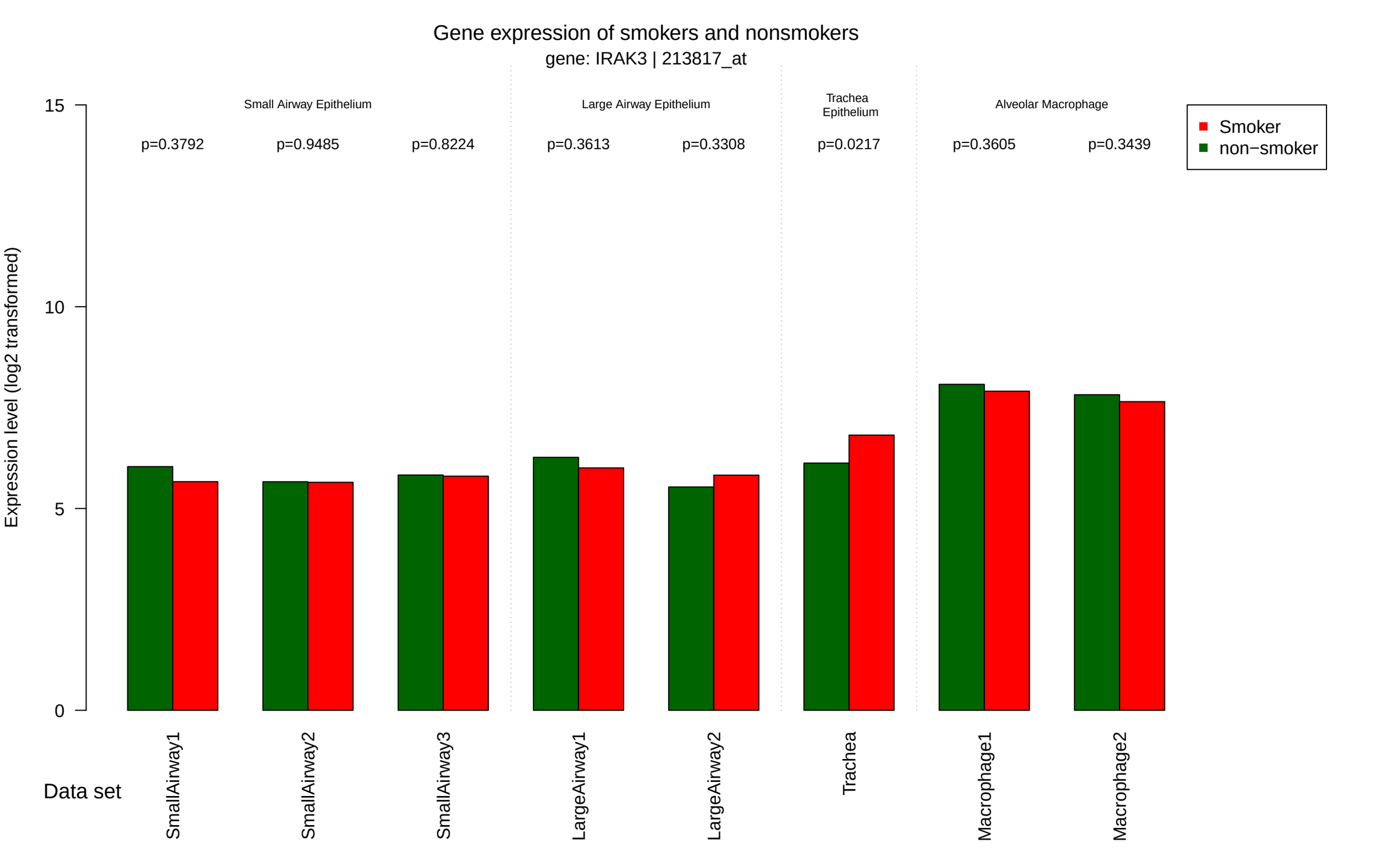
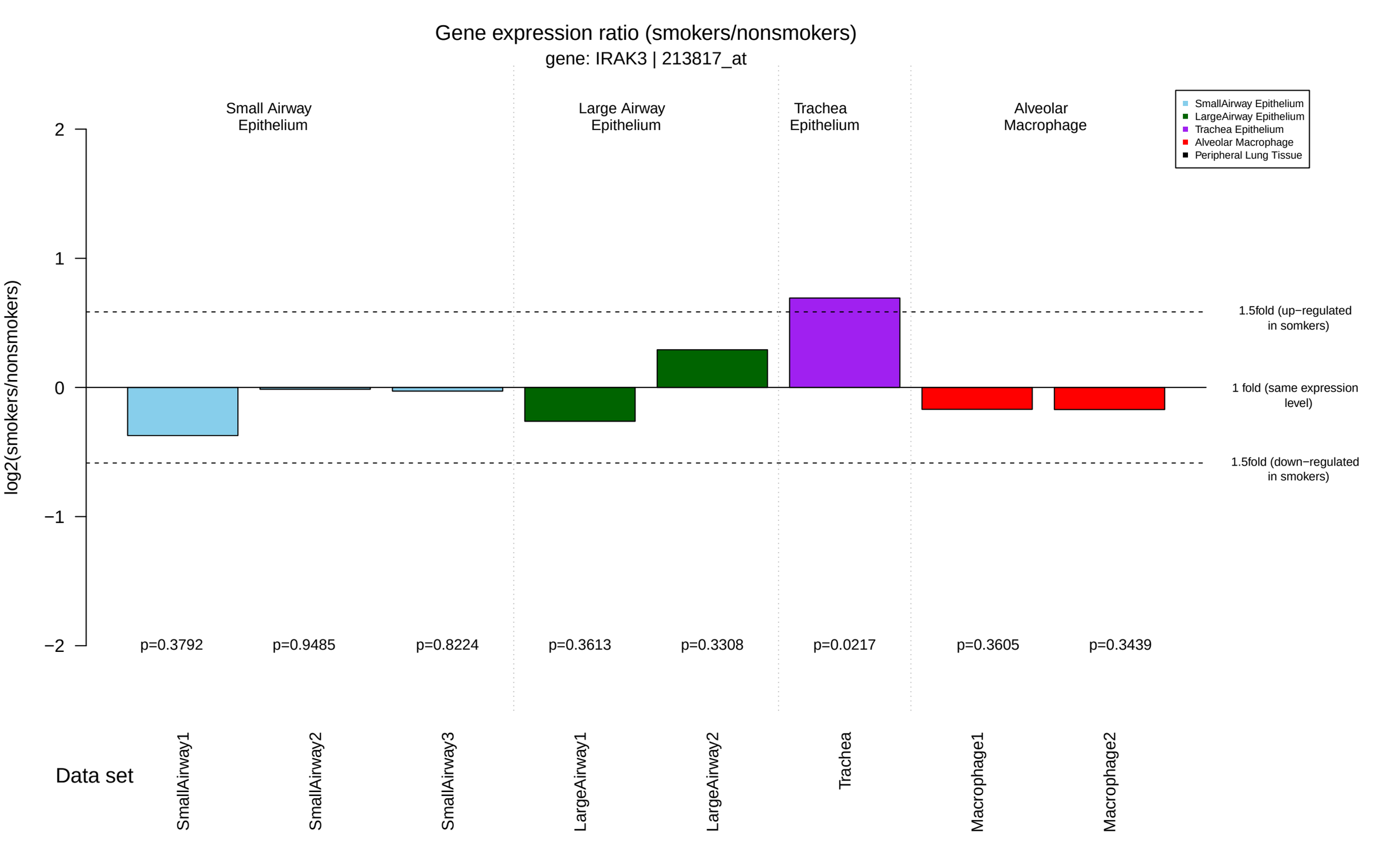
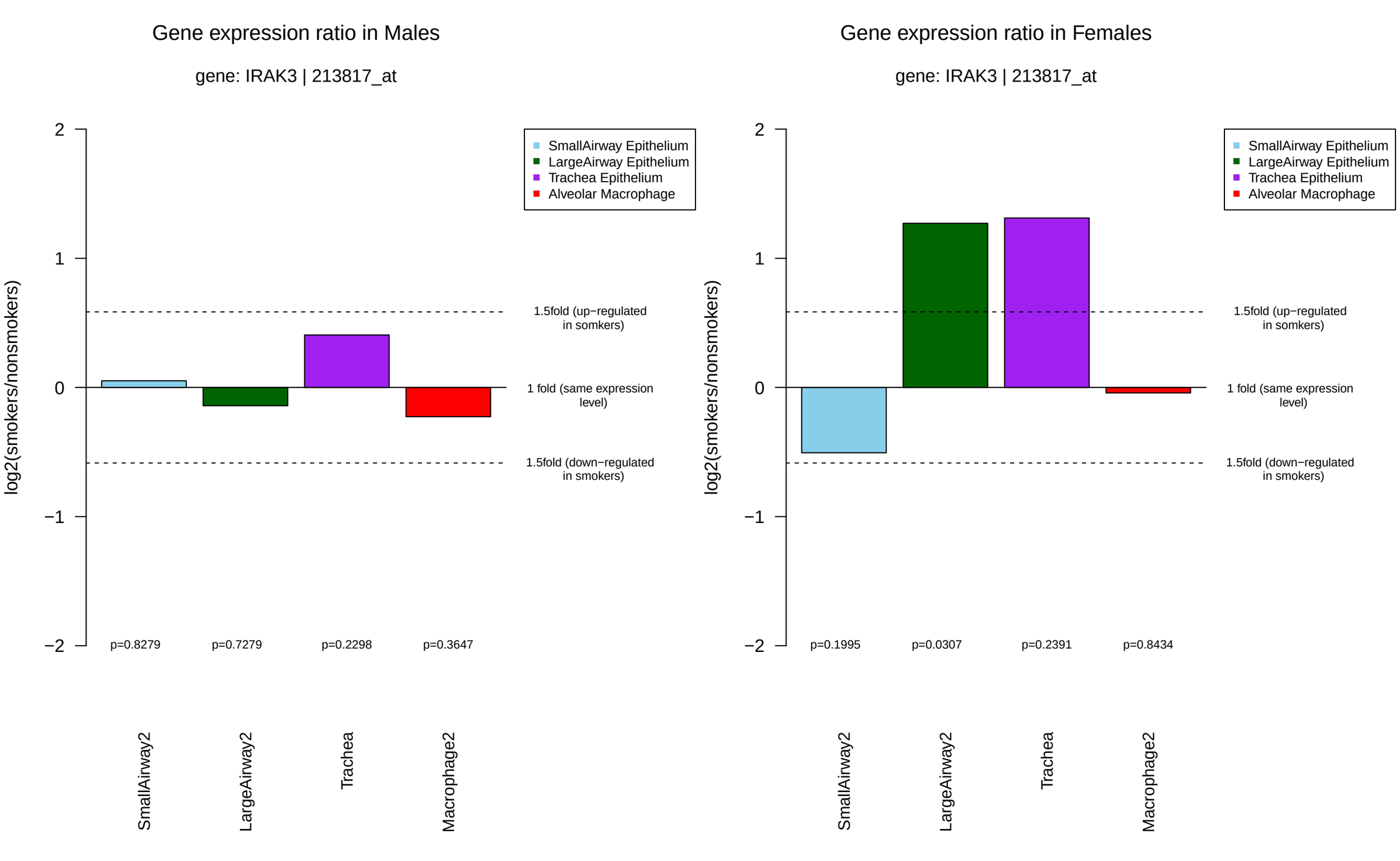
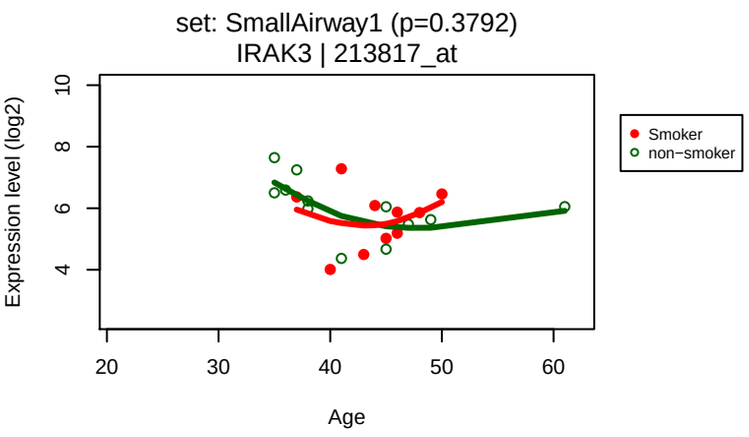
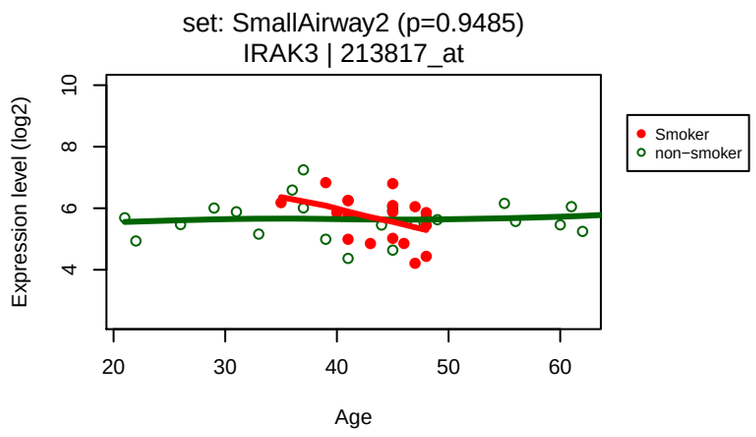
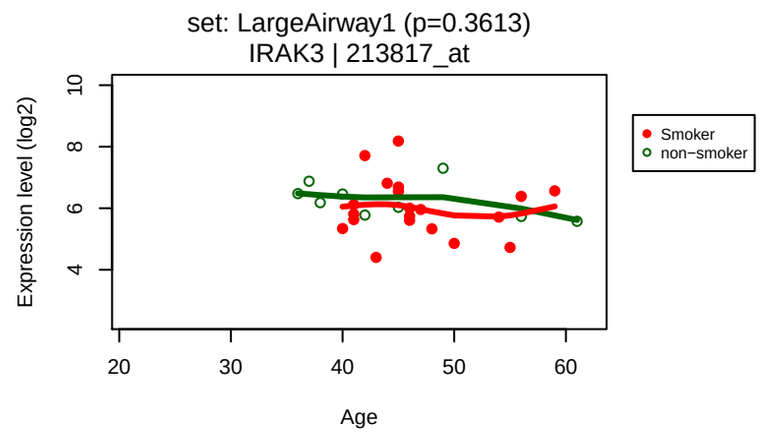

Gene expression of smokers and nonsmokers
gene: IRAK3 | 213817_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3792
p=0.9485
p=0.8224
p=0.3613
p=0.3308
p=0.0217
p=0.3605
p=0.3439
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

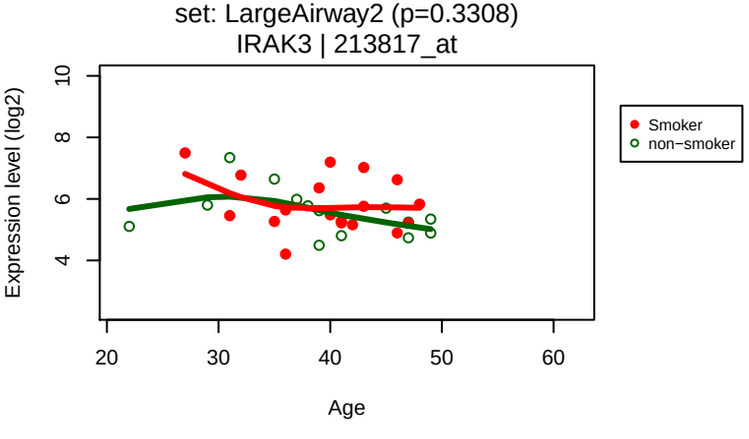

Gene expression ratio (smokers/nonsmokers)
gene: IRAK3 | 213817_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3792
p=0.9485
p=0.8224
p=0.3613
p=0.3308
p=0.0217
p=0.3605
p=0.3439
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

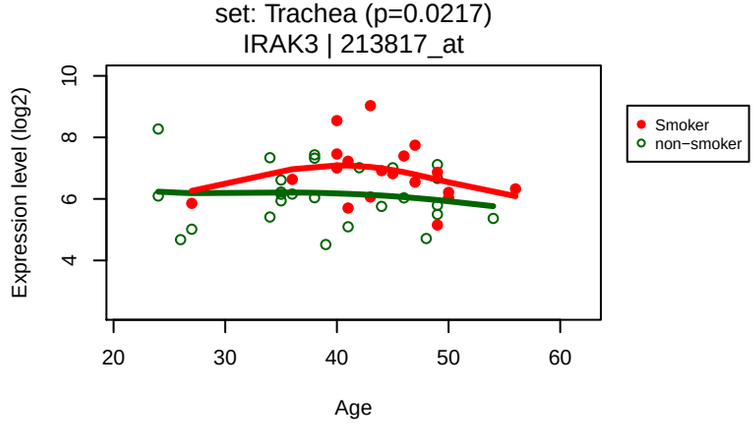

Gene expression ratio in Males
gene: IRAK3 | 213817_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8279
p=0.7279
p=0.2298
p=0.3647
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRAK3 | 213817_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1995
p=0.0307
p=0.2391
p=0.8434
SmallAirway2
LargeAirway2
Trachea
Macrophage2

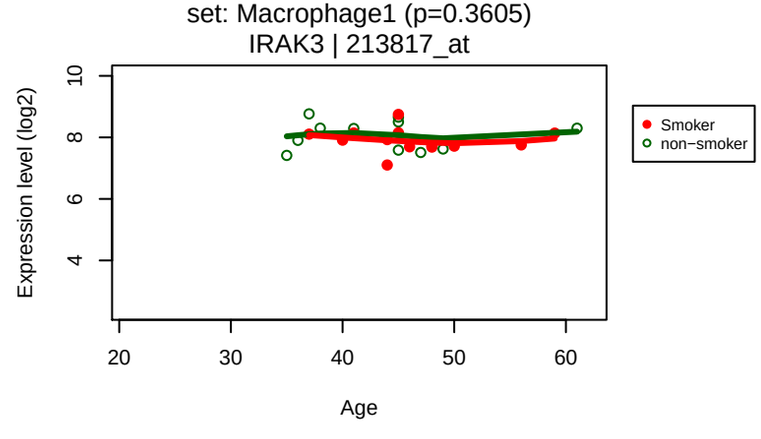

set: SmallAirway1 (p=0.3792)
IRAK3 | 213817_at
Expression level (log2)
Age
Smoker
non−smoker

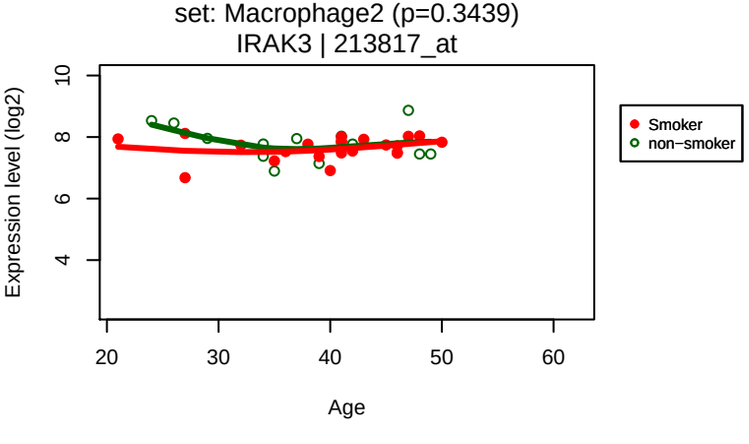

set: SmallAirway2 (p=0.9485)
IRAK3 | 213817_at
Expression level (log2)
Age
Smoker
non−smoker

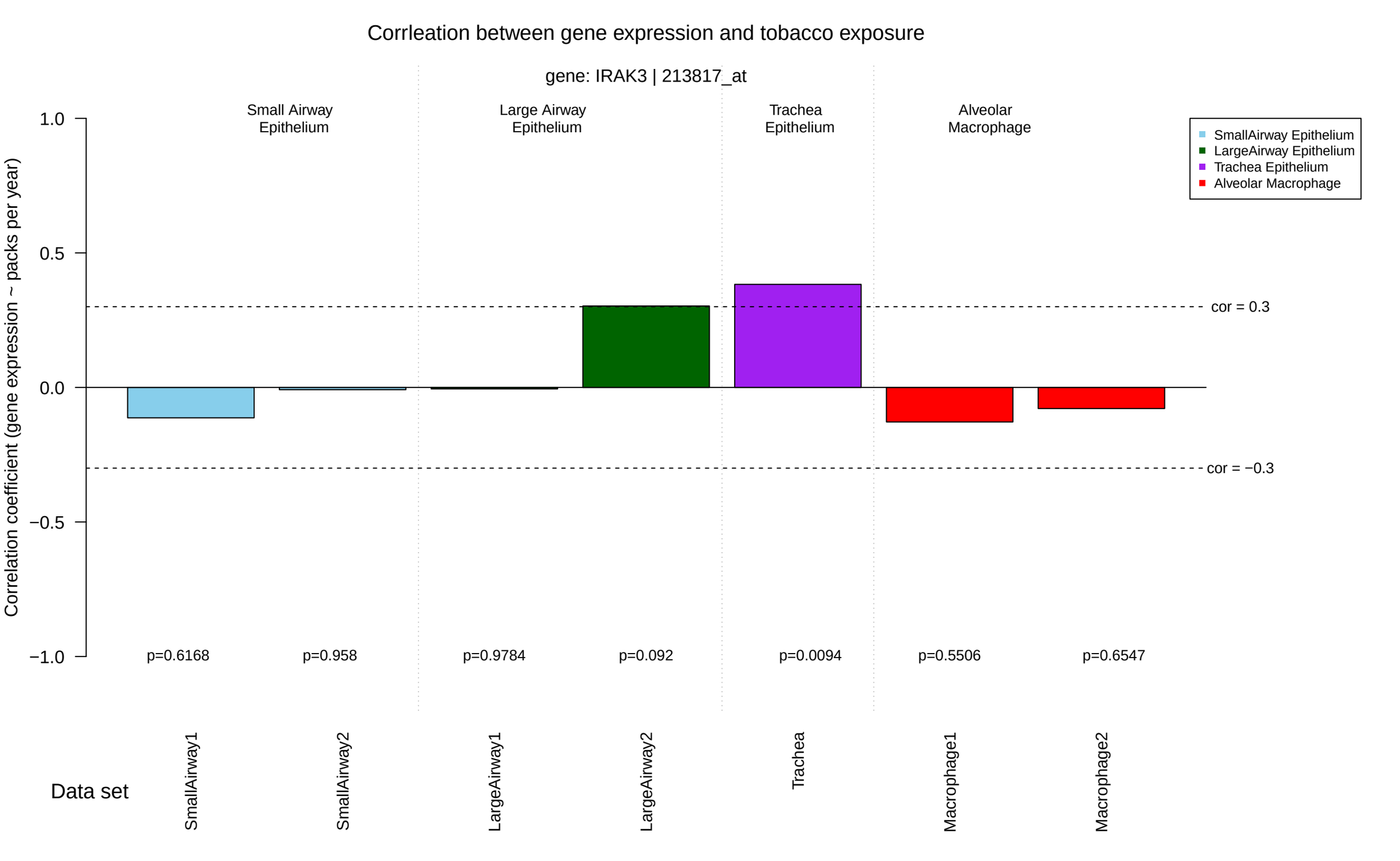

set: LargeAirway1 (p=0.3613)
IRAK3 | 213817_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3308)
IRAK3 | 213817_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0217)
IRAK3 | 213817_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3605)
IRAK3 | 213817_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3439)
IRAK3 | 213817_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRAK3 | 213817_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6168
p=0.958
p=0.9784
p=0.092
p=0.0094
p=0.5506
p=0.6547
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2